

NHS GP Appointment Data Analysis

Technical Report | DA201 Assignment

Student : Sharath. Sista | 2026, May | Audience: Technical Stakeholders

742.8M

Total Appointments

30.9M**(4.2%)**

DNA Appointments

GBP371M

Est. DNA Cost/Year

42 ICBs

ICBs Covered

Jan 2020**to Jun****2022**

Date Range

1. Background and Business Context

The NHS faces growing demand for GP services alongside tight budget constraints. This analysis was commissioned to answer two questions: Has there been adequate staff and capacity within the networks, and what has been the actual utilisation of resources? The central focus is missed appointments, known as Did Not Attend (DNA), which cost the NHS approximately 371 million pounds per year at 30 pounds per missed appointment.

The BMA chair, Professor Philip Banfield, has stated that the reasons patients miss appointments should be investigated rather than responded to with penalties, as financial penalties disproportionately affect the poorest communities. This report presents data-driven evidence to support that investigative approach and argues that this is an operational problem, not a capital one.

2. Analytical Approach

2.1 Data Sources

File	Description	Rows	Date Range
actual_duration.csv	Appointment duration at sub-ICB level	137,793	Dec 2021 to Jun 2022
appointments_regional.csv	Status, mode, HCP type, lead time by ICB	596,821	Jan 2020 to Jun 2022
national_categories.xlsx	Service setting, context type, clinical category	817,394	Aug 2021 to Jun 2022
tweets.csv	NHS-related Twitter data	1,174	N/A

2.2 Data Quality Notes

All datasets were imported using the pandas library in Python. The count_of_appointments column is a sum of appointment numbers, not a row count. Totals are estimates because not all GP practices are included. Key limitations noted and documented throughout the notebook:

- Unknown appointment mode: Cegedim system practices cannot supply mode data, increasing Unknown entries from July 2019
- Unmapped categories: reflect gaps in NHS GP recording systems designed for clinical use, not analysis
- HCP type: only the GP category is reliably captured due to known extraction errors from October 2017
- Duration data covers December 2021 to June 2022 only and excludes durations under one minute or over 60 minutes

2.3 Wrangling and Analysis Method

Date columns were converted to datetime objects using `pd.to_datetime()` with explicit format strings. For `national_categories.xlsx`, `appointment_month` was derived from `appointment_date` using `.dt.to_period()` to avoid timezone conflicts in newer versions of pandas. All Unknown and Unmapped entries were retained and documented. DNA rates were calculated as DNA appointments divided by total appointments, grouped across four dimensions: mode, booking lead time, HCP type, and ICB region.

3. Key Findings from the Analysis

Appointment status split: Of 742.8 million total appointments, 91.2% were attended, 4.16% were DNA representing 30.9 million missed appointments, and 4.6% had Unknown status. At 30 pounds per missed appointment, the estimated cost over 30 months is 927 million pounds, or 371 million pounds per year.

Monthly trend: Appointments fell by over 40% during COVID-19 lockdowns in April and May 2020, recovering to pre-pandemic levels by late 2020. Demand continued rising through 2021 and 2022, suggesting increasing pressure on the system.

Telephone surge during COVID: Telephone appointments increased by 151%, from 3.5 million to 8.8 million per month, with zero new infrastructure. This proves the NHS can scale operationally without capital investment when needed.

DNA by booking lead time (key finding): Same-day appointments have a DNA rate of 1.77%. Appointments booked 28 or more days ahead have a DNA rate of 8.83%, nearly five times higher. Patients are not choosing to wait this long. They wait because no earlier slots are available.

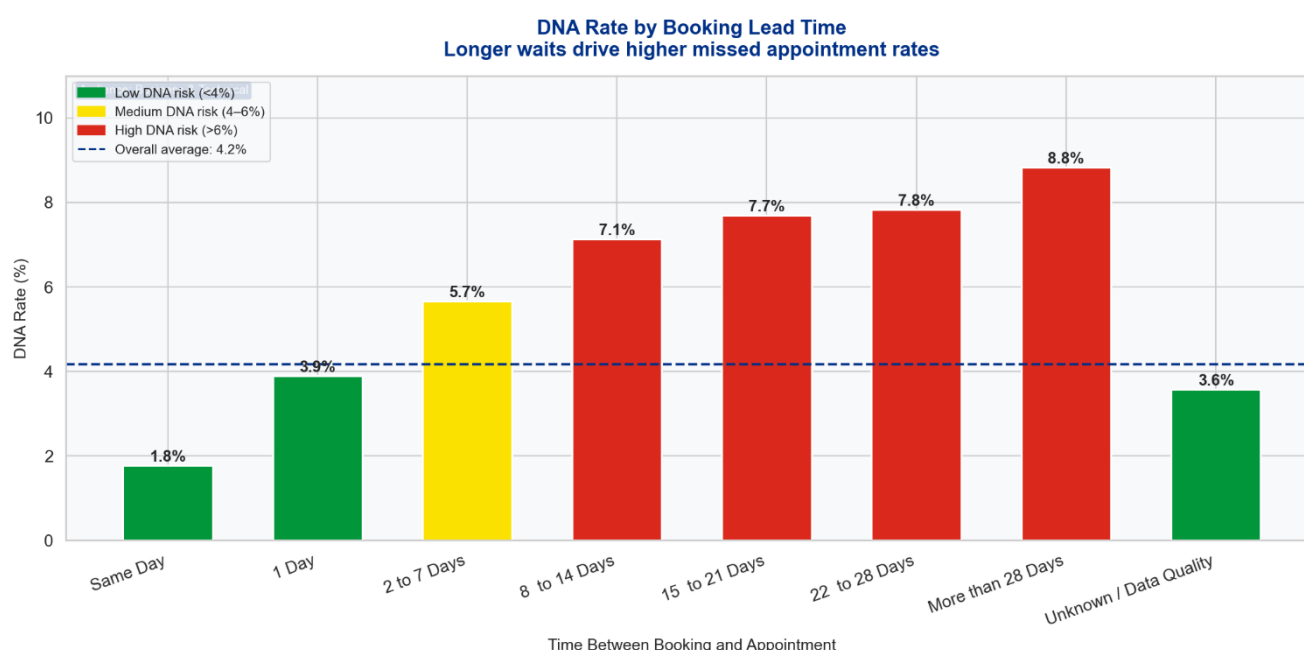


Figure 1: Escalation of NHS Did Not Attend (DNA) Rates by Scheduling Lead Time Horizon.

DNA by mode: Telephone appointments have a DNA rate of 1.96% compared to 5.51% for face-to-face, a difference of 2.8 times. Telephone already accounts for 268 million appointments and is the most reliable mode available.

Regional variation: DNA rates range from 2.69% to 5.84% across the 42 ICBs, a gap of 3.15 percentage points. These ICBs share the same national infrastructure. The difference is in how they operate, not what they have built.

ICB regional performance analysis: A composite performance score was built for all 42 ICBs using four weighted metrics: DNA rate (40%), same-day booking rate (30%), telephone utilisation rate (20%), and unknown status rate (10%). Northamptonshire ranked first with a score of 82.7 and a DNA rate of 2.97%. Greater Manchester ranked last with a score of 27.2 and a DNA rate of 5.71%. The 55.5 point gap between best and worst, using identical national infrastructure, is the strongest evidence that this is an operational problem not a capital one. If Greater Manchester alone matched Northamptonshire DNA rate, 933,808 appointments would be recovered and approximately 28 million pounds saved from that one ICB

alone. A quadrant analysis of DNA rate versus same-day booking rate further classified each ICB by problem type, with those showing high DNA and low same-day rates needing the most urgent operational intervention.

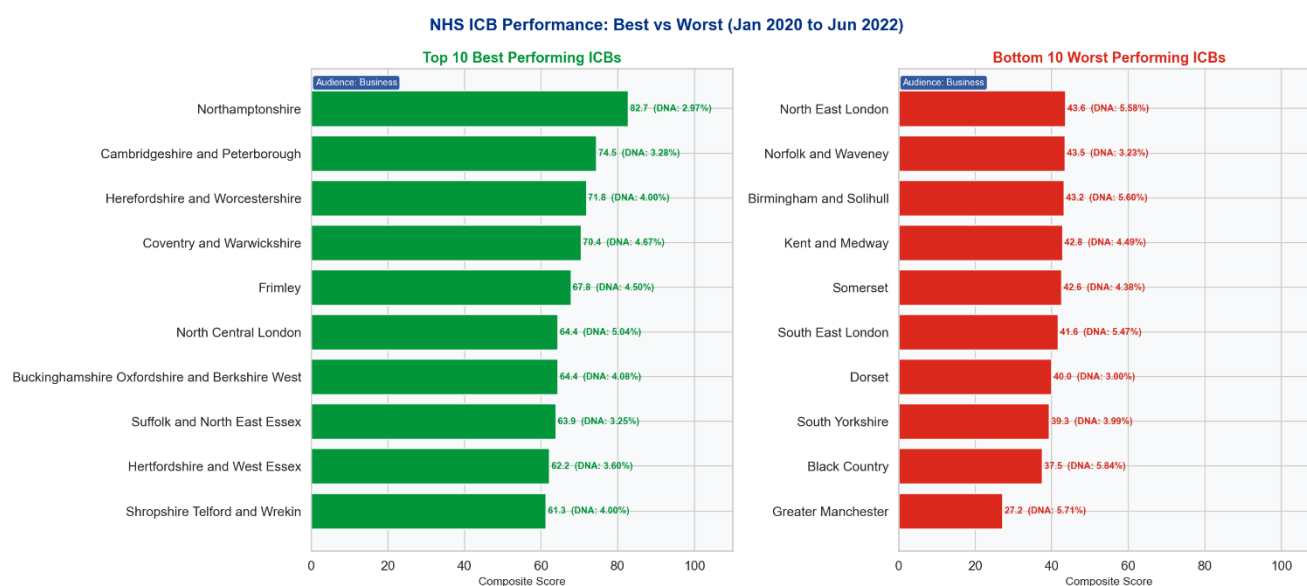


Figure 2: Multi-Year Longitudinal Analysis of Monthly NHS GP Did Not Attend (DNA) Volumetric and Percentage Rates (Jan 2020 – Jun 2022).

Twitter analysis: The top hashtags were NHS, Healthcare, and MentalHealth. Mental health appears prominently in public discourse, signalling a concern not fully visible in appointment data. Workforce tags such as Nurses and Hiring also featured. Twitter data should supplement rather than replace primary analysis.

4. Answering the Two Business Questions

Question 1: Has there been adequate staff and capacity?

Answer: The physical capacity broadly exists. The operational use of that capacity does not.

Three pieces of evidence prove this is an operational problem rather than a capital one. First, 46.1% of all appointments are booked same-day and have a DNA rate of just 1.77%. If buildings and rooms were the bottleneck, same-day appointments at this scale would not be possible. Second, telephone appointments increased by 151% during COVID with zero new infrastructure, proving the system can flex operationally without capital investment. Third, DNA rates vary by 3.15 percentage points across ICBs using identical national infrastructure. The same buildings and systems produce very different results, which is an operational gap not a capital one.

The vicious cycle at work: limited slots force patients to book far ahead, long-wait bookings generate DNA rates nearly five times higher than same-day bookings, wasted slots reduce effective capacity further, and the cycle repeats. Breaking this cycle requires operational change, not new buildings.

The Capacity-DNA Vicious Cycle



Figure 3: Empirical Breakdown of the NHS Capacity-DNA Vicious Cycle and Booking Volume-to-Reliability Corridors.

Question 2: What was the actual utilisation of resources?

Answer: 91.2% of booked capacity was attended, but the 8.8% lost to DNA and Unknown status is concentrated in preventable areas.

Face-to-face accounts for 59.2% of attended appointments but has the highest DNA rate at 5.51%. Telephone has a DNA rate of just 1.96% and already handles 268 million appointments. Video and online appointments account for only 0.49% despite being proven at scale during COVID. If all modes matched the telephone DNA rate, the NHS would recover 16.35 million appointments over 30 months, worth approximately 196 million pounds per year. That saving requires no capital expenditure at all.

5. Recommendations

Investment should follow an OpEx-first sequence. Fix operational problems, measure the remaining gap, then assess whether capital expenditure is needed.

Priority	Type	Action	Evidence	Impact
1	OpEx	Reduce forward booking time to same-day or next-day	DNA 1.77% (same-day) vs 8.83% (28+ days)	GBP19.5M/yr
2	OpEx	Expand telephone for routine and follow-up care	Telephone DNA 1.96% vs face-to-face 5.51%	GBP196M/yr
3	OpEx	SMS reminders 24 to 48 hours before appointments	Sussex ICB reduced DNA from 6.5% to 5.5%	GBP15-30M/yr
4	OpEx	Target high-DNA ICBs with operational review	3.15pp gap across 42 ICBs same infrastructure	GBP50M+ est.

5	OpEx	Link DNA to deprivation index for equity targeting	Most deprived patients 10 to 20% more likely to miss	Equity focus
6	CapEx	Scale video and online appointment infrastructure	Only 0.49% of appointments, proven during COVID	Medium-term
7	CapEx	New GP surgery capacity if OpEx gap persists	Reassess after 12 to 24 months of OpEx improvements	Deferred

6. Further Analysis Recommended

- Link sub-ICB DNA rates to the Index of Multiple Deprivation via UK LLC NHS England datasets to prove the deprivation-DNA relationship and enable equity-focused targeting
- Correlate booking lead time with DNA rates by ICB to identify which ICBs have both the longest waits and the highest DNA rates, confirming the booking system as the primary lever
- Model projected savings from a phased 12 to 24 month OpEx improvement programme across all 42 ICBs
- Extend duration analysis beyond December 2021 to June 2022 for seasonal capacity planning
- Apply sentiment analysis to Twitter data beyond hashtag counting for richer public signal

Word count: approximately 1,200 words | Notebook: Sharath_Sista_DA201_Assignment_Notebook.ipynb